Exploratory Data Analysis of the SIIM-ISIC Melanoma Classification Dataset

1. Introduction

Melanoma is a kind of skin cancer that starts in the melanocytes - the cells responsible for producing pigment in the skin. The pigment is called melanin.

The exact cause of all melanomas isn't clear. Although melanoma is less common than some other forms of skin cancer, it accounts for a significant proportion of skin cancer-related deaths. Early and accurate detection is therefore an important area of research, and computer vision and machine learning techniques have the potential to assist medical professionals in the identification of suspicious skin lesions.

The purpose of this exploratory analysis is to gain a better understanding of the dataset before applying machine learning / deep learning techniques. The analysis examines the structure and quality of the metadata, the distribution of the target variable, patient-level information, and characteristics of the available images. Particular attention is given to missing values, duplicate observations and other factors that may influence the performance and reliability of a predictive model.

2. Dataset Description

Dataset size

Train dataset size	33126 x 8
Test dataset size	10982 x 5

Train and test structure

Train	image_name	patient_id	sex	age_approx	anatom_site_general_challenge	diagnosis	benign_malignant	target
Test	image_name	patient_id	sex	age_approx	anatom_site_general_challenge	-	-	-

Missing values

Train dataset	Missing values	Test dataset	Missing values
image_name	0	image_name	0
patient_id	0	patient_id	0
sex	65	sex	0
age_approx	68	age_approx	0
anatom_site_general_challenge	527	anatom_site_general_challenge	351
diagnosis	0	-	
benign_malignant	0	-	
target	0	-	

The missing values in **sex** and **age_approx** are **dependent / correlated**.

Target distribution

Malignant rate	1.8%
Benign rate	98.2%

The target distribution is highly imbalanced. This must be addressed during the training phase.
Possible solutions -> **data augmentation,**
resampling,
class weighting

Duplicates

There were no duplicate rows found in either dataset. However, a large number of patients have multiple entries, which means the data is not independent at the row level. Visual inspection confirmed that these repeated entries correspond to different lesions or imaging angles rather than duplicate photos, but this still introduces a **risk of data leakage**, as a single patient's images may share underlying characteristics (e.g. skin tone or mole color).
As a result, randomly splitting the dataset by row could allow information about a specific patient to leak between training and validation sets. A patient-level split is recommended during preprocessing.
Below is the analysis that shows that 33,126 images belong to only 2,056 patients.

Unique values per column

Column name	Unique values
image_name	33126
patient_id	2056
sex	2
age_approx	18
anatom_site_general_challenge	6
diagnosis	9
benign_malignant	2
target	2

Outliers: age = 0

Found only 2 rows with the same patient who also has an age = 10 entry.
Possibilities: data entry typo or a 10-year gap between visits

Image resolution

Image resolutions vary across the datasets, ranging from 640x480 up to 6000x4000. Therefore all images must be **resized** to a consistent input size during the preprocessing phase.

Feature distributions

Sex: 51.7% male, 48.3% female; malignant rate differs slightly
Age: with skewness = 0.081, the distribution is almost symmetric; malignant rate increases with age
Anatomical site: torso has the highest sample count in the dataset but a moderate malignant rate; sites with fewer samples like oral/genital show less reliable rate estimates and should be interpreted with caution

3. Conclusion for the next steps

1. **Missing data**
   The proportion of missing data is very low relative to the dataset size (sex - 0.196%, age_approx - 0.205%, anatom_site_general_challenge - 1.59% in the training dataset and anatom_site_general_challenge - 3.196% in the test dataset), so dropping the affected rows would have minimal impact on the overall dataset.
   Alternatively, sex and anatom_site_general_challenge can be filled using the mode or an "unknown" category, and age_approx can be filled using the median.
2. **Class imbalance**
   As the data is highly imbalanced, we can use data augmentation, resampling or class weighting (downsampling will cause a huge data loss)
3. **Image preprocessing**
   As noted in the image resolution analysis, all images must be resized to a consistent input size before being used for model training. Additionally, removing hair artifacts from images may further improve model accuracy by reducing visual noise unrelated to the lesion itself.
   Source - https://www.mdpi.com/2076-3417/16/4/1819
4. **A patient-level split**
   Given that a large number of patients have multiple image entries, a patient-level split may be used when creating training and validation sets to prevent information about a specific patient from leaking between them.
5. **Encoding**
   image_name and patient_id are identifier columns rather than predictive features. image_name is required to load the corresponding image file for each observation, while patient_id may be used to define patient-level train/validation splits. The remaining categorical columns (sex, anatom_site_general_challenge, diagnosis, benign_malignant) need to be encoded into a numeric format before being used in a model (e.g. OHE).